

PetPal — Resources & Attribution

FBLA Introduction to Programming 2025-2026 Developer: Tharun Naguleswaran

External Libraries & APIs

These are the only external dependencies loaded by the project. All other code is original.

Resource	Version / Model	Source	Purpose	License
Chart.js	v4.4.4	cdn.jsdelivr.net	Renders all charts in the Reports tab (line, bar, doughnut)	MIT
Press Start 2P	—	Google Fonts (fonts.googleapis.com)	Pixel-style heading and UI font	OFL (SIL Open Font License)
Pixelify Sans	400–700	Google Fonts (fonts.googleapis.com)	Secondary pixel display font	OFL (SIL Open Font License)
Groq API	llama-3.1-8b-instant	api.groq.com	Powers the Dr. Paws AI chat advisor in the Chat tab	Groq Terms of Service

Chart.js

- **URL:** <https://cdn.jsdelivr.net/npm/chart.js@4.4.4/dist/chart.umd.min.js>
- **License:** MIT — <https://github.com/chartjs/Chart.js/blob/master/LICENSE.md>
- **Usage:** All four report charts — expense doughnut, monthly bar, health line chart, achievement stacked bar, and care history bar

Press Start 2P

- **URL:** <https://fonts.google.com/specimen/Press+Start+2P>
- **Designer:** CodeMan38
- **License:** SIL Open Font License 1.1
- **Usage:** Primary display font throughout the UI (`--font-pixel` CSS variable)

Pixelify Sans

- **URL:** <https://fonts.google.com/specimen/Pixelify+Sans>
- **Designer:** Stefie Justprince
- **License:** SIL Open Font License 1.1
- **Usage:** Secondary pixel font for body text and labels

Groq API

- **Endpoint:** <https://api.groq.com/openai/v1/chat/completions>

- **Model:** llama-3.1-8b-instant (Meta, open weights)
- **Terms:** <https://groq.com/terms-of-service/>
- **Usage:** Streaming AI chat responses for the Dr. Paws advisor in the Chat tab. Requires a free user-supplied API key — no key is bundled with the project.

Code

File	Description
js/constants.js	All game configuration, costs, stat definitions, badge/trick/chore data
js/icons.js	Custom pixel SVG icon library (all icons hand-authored)
js/PixelSprites.js	Animated pixel art pet renderer using the HTML5 Canvas API
js/Validator.js	Centralized input validation (syntactical and semantic)
js/Pet.js	Pet model — stats, XP, leveling, tick decay, personality effects
js/StorageManager.js	localStorage persistence layer for game state, expenses, and income
js/Game.js	Game coordinator — actions, chores, badges, save/load
js/AICompanion.js	Groq API integration with streaming response handling
js/PersonalityEngine.js	Adaptive personality trait system
js/Scrapbook.js	Auto-recorded milestone memory journal
js/ReportsCenter.js	Chart.js analytics with date filtering and downloadable reports
js/HelpCenter.js	Searchable in-game FAQ
js/UIController.js	All DOM interaction, game loop, and tab management
js/main.js	Application entry point
css/styles.css	Main stylesheet
css/animations.css	Keyframe animations
index.html	Single-page application structure

Game Concept Inspiration

The gameplay mechanics were inspired by established virtual pet games. No code or assets were taken from any of these sources — only the general concept of stat-based pet care.

Inspiration	What Was Referenced
Tamagotchi (Bandai, 1996)	Core concept of time based stat decay and care actions
Neopets (1999)	In-game currency and financial management mechanics
Pokémon series (Nintendo/Game Freak)	Emotion and personality systems

Development Tools

These tools were used during development but are not part of the submitted project.

Tool	Purpose
Visual Studio Code	Code editor
Google Chrome DevTools	Debugging and testing
Git / GitHub	Version control
MDN Web Docs (developer.mozilla.org)	JavaScript and Web API reference

Web Standards Referenced

- **HTML5** — <https://html.spec.whatwg.org/>
- **CSS3 / CSS Custom Properties** — <https://www.w3.org/Style/CSS/>
- **ES6+ JavaScript** — <https://www.ecma-international.org/>
- **Web Storage API (localStorage)** — <https://developer.mozilla.org/en-US/docs/Web/API/Window/localStorage>
- **Canvas API** — https://developer.mozilla.org/en-US/docs/Web/API/Canvas_API
- **Fetch API** — https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API
- **ARIA / Accessibility** — <https://www.w3.org/WAI/ARIA/apg/>

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